Failure Analysis Report

FAIL-0914 — Data Mismatch

1. Failure Summary

Failure ID	FAIL-0914
Failure Type	Data Mismatch
Severity	Minor
Component	Database
Application	APP-010 — MediaPlatform
Build Version	5.1.9
Timestamp	2023-07-27 06:26:13
Environment	ENV-007 — Staging
OS Version	Android 13
Network	LTE
Resolution	Resolved

2. Test Case Reference

Test Case ID	TC-1465
Title	Ensure background sync synchronizes correctly across devices when the user revok...
Execution ID	EXEC-04860
Test Run	RUN-011 — Regression Cycle 11 — Build 6.3.9
Requirement	REQ-0325 — Functional — Secure Key Rotation

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete firmware over-the-air update within the defined SLA.
Observed Behaviour	The system exhibited data mismatch when the user revokes permissions during the operation.

4. Error Details

Error Message:

```
DataIntegrityError: field 'deviceId' expected=true actual=disconnected.
```

Stack Trace / Log Excerpt:

```
com.samsung.health.Authentication.execute(line 401)
```

Log Reference	LOG-84019

5. Executed Steps at Time of Failure

SYNTHETIC DATA — No real Samsung confidential information

1	Enter valid credentials: username='testuser_005', password='[REDA...	The operation completes without error.
2	Re-connect the headset and confirm auto-recovery.	The UI transitions to the next screen within 3 seconds.
3	Simulate network drop using the network emulator profile 'Offline...	No crash or ANR is observed.
4	Verify that the UI displays 'Connection successful'.	The performance counter logs a value below the SLA threshold.
5	Check server logs for event code 'EVT-2278'.	The audit log records the event with correct metadata.
6	Navigate to Settings > Account and tap 'Pair Device'.	The device reconnects automatically within 10 seconds.

6. Root Cause Analysis

Missing null-check on the device registry response before dereferencing the session token.

8. Historical Similar Failures

HIST-0303	FAIL-0304	Data Mismatch	5.0.7	2024-01-16	2	Wont Fix
HIST-0608	FAIL-0609	Data Mismatch	3.5.3	2021-08-24	14	Wont Fix
HIST-0913	FAIL-0914	Data Mismatch	4.3.3	2021-04-14	9	Wont Fix

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

SYNTHETIC DATA — No real Samsung confidential information